

## CHITTAGONG SHAH AMANAT INTL, Bangladesh

WMO: 419780

Lat: 22.250N

Lon: 91.813E

Elev: 12

StdP: 14.69

Time Zone: 6.00 (E06)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 57.3       | 59.4 | 46.6                          | 47.0 | 75.4 | 50.0 | 53.3 | 73.4 | 17.8                  | 75.3 | 15.8 | 75.8 | 2.1                   | 20   | 0.333 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 10.9                   | 93.1            | 81.4 | 91.7 | 81.0 | 91.0 | 80.7 | 83.9                | 89.8 | 83.2 | 88.8 | 82.5 | 87.9 | 9.5                  | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 82.6                            | 169.9 | 87.6 | 81.5 | 164.1 | 86.9 | 80.8 | 160.2 | 86.5 | 48.3          | 89.7 | 47.3 | 89.3 | 46.6 | 88.6 | 93.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |     |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|-----|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |     |     |
| (a)               | (b)  | (c)  | (d)                        | (e) | (f)                | (g) | (h)                                                | (i) | (j)        | (k) | (l)        | (m) | (n)        | (o) | (p) | (q) |
| 18.4              | 15.7 | 13.7 | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A | N/A | N/A |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A | N/A | N/A |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 79.5   | 68.8  | 73.4  | 79.4  | 82.9  | 84.0  | 84.0  | 83.0  | 83.3  | 83.6  | 82.4  | 77.7  | 71.2  |
|                                                                        | DBStd       | 5.89   | 2.98  | 3.06  | 2.49  | 2.59  | 2.81  | 2.58  | 2.15  | 2.00  | 1.87  | 2.57  | 3.08  | 3.68  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 9      | 5     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     |
|                                                                        | CDD50       | 10767  | 581   | 656   | 911   | 988   | 1053  | 1021  | 1025  | 1033  | 1007  | 1003  | 831   | 658   |
|                                                                        | CDD65       | 5302   | 122   | 236   | 446   | 538   | 588   | 571   | 560   | 568   | 557   | 538   | 381   | 197   |
|                                                                        | CDH74       | 56229  | 520   | 1474  | 4179  | 6419  | 7323  | 6938  | 6381  | 6607  | 6555  | 5869  | 3028  | 936   |
|                                                                        | CDH80       | 20123  | 35    | 274   | 1291  | 2590  | 3277  | 2825  | 2142  | 2319  | 2405  | 2061  | 794   | 110   |
| (14) Wind                                                              | WSAvg       | 6.0    | 5.3   | 5.4   | 6.0   | 7.9   | 7.4   | 7.6   | 8.2   | 7.1   | 5.7   | 3.7   | 3.7   | 3.8   |
| (15) Precipitation                                                     | PrecAvg     | 116.0  | 0.2   | 0.9   | 2.5   | 4.1   | 13.9  | 25.1  | 26.4  | 19.3  | 12.4  | 9.6   | 1.7   | 0.4   |
|                                                                        | PrecMax     | 155.4  | 2.1   | 4.3   | 8.9   | 11.3  | 30.2  | 50.2  | 55.9  | 45.2  | 28.7  | 24.3  | 11.8  | 3.3   |
|                                                                        | PrecMin     | 88.6   | 0.0   | 0.0   | 0.0   | 0.0   | 4.3   | 7.9   | 11.3  | 6.1   | 5.6   | 2.5   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 19.8   | 0.5   | 1.3   | 2.5   | 3.1   | 7.1   | 10.8  | 11.1  | 8.5   | 6.0   | 5.3   | 2.5   | 0.8   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 84.1  | 87.8  | 91.7  | 94.3  | 94.5  | 93.6  | 91.7  | 92.2  | 93.1  | 92.3  | 89.8  | 84.6  |
|                                                                        |             | MCWB   | 67.8  | 66.6  | 70.4  | 79.6  | 82.7  | 82.8  | 82.1  | 81.8  | 81.9  | 80.6  | 76.1  | 69.7  |
|                                                                        | 2%          | DB     | 80.7  | 85.3  | 89.5  | 91.4  | 92.9  | 92.0  | 89.9  | 91.0  | 91.2  | 91.0  | 87.8  | 82.7  |
|                                                                        |             | MCWB   | 65.6  | 67.6  | 72.6  | 80.1  | 82.0  | 82.6  | 81.4  | 81.5  | 81.4  | 79.7  | 75.0  | 68.7  |
|                                                                        | 5%          | DB     | 79.0  | 83.4  | 87.7  | 89.7  | 91.4  | 91.1  | 88.6  | 89.4  | 89.6  | 89.5  | 86.1  | 80.9  |
|                                                                        |             | MCWB   | 64.6  | 67.7  | 73.7  | 79.1  | 81.4  | 82.2  | 80.9  | 80.9  | 80.6  | 79.2  | 74.1  | 68.4  |
|                                                                        | 10%         | DB     | 77.2  | 81.9  | 86.0  | 88.2  | 89.9  | 89.6  | 87.5  | 87.7  | 87.9  | 87.9  | 84.4  | 79.1  |
|                                                                        |             | MCWB   | 64.5  | 67.6  | 73.7  | 78.7  | 80.7  | 81.5  | 80.5  | 80.5  | 80.3  | 78.7  | 72.6  | 67.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 71.7  | 75.6  | 80.1  | 83.5  | 85.0  | 85.2  | 84.6  | 83.6  | 83.6  | 82.8  | 79.6  | 74.7  |
|                                                                        |             | MCDB   | 77.7  | 80.6  | 85.7  | 90.2  | 91.6  | 90.3  | 88.7  | 89.3  | 89.1  | 89.0  | 85.1  | 78.5  |
|                                                                        | 2%          | WB     | 69.4  | 74.0  | 79.1  | 82.1  | 83.6  | 84.0  | 82.8  | 82.5  | 82.5  | 81.6  | 78.1  | 72.0  |
|                                                                        |             | MCDB   | 75.4  | 78.8  | 84.5  | 88.0  | 89.9  | 89.6  | 87.3  | 87.4  | 88.1  | 87.5  | 83.6  | 78.6  |
|                                                                        | 5%          | WB     | 67.6  | 72.6  | 78.2  | 81.2  | 82.7  | 83.1  | 81.8  | 81.7  | 81.7  | 80.7  | 77.0  | 70.4  |
|                                                                        |             | MCDB   | 74.7  | 77.7  | 83.5  | 86.9  | 88.4  | 88.7  | 86.4  | 86.5  | 87.0  | 86.5  | 82.7  | 77.3  |
|                                                                        | 10%         | WB     | 66.3  | 71.2  | 77.2  | 80.2  | 81.8  | 82.3  | 80.9  | 81.0  | 80.8  | 79.9  | 75.9  | 69.4  |
|                                                                        |             | MCDB   | 74.3  | 77.1  | 82.5  | 85.9  | 87.5  | 87.8  | 85.4  | 85.7  | 86.0  | 85.4  | 81.5  | 76.2  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.5  | 17.0  | 14.1  | 11.5  | 10.9  | 9.3   | 7.7   | 8.3   | 8.8   | 10.4  | 13.9  | 15.5  |
|                                                                        |             | MCDBR  | 19.2  | 18.7  | 16.4  | 12.8  | 12.6  | 11.8  | 9.9   | 10.8  | 11.3  | 12.5  | 14.4  | 16.8  |
|                                                                        |             | MCWBR  | 8.3   | 7.6   | 7.6   | 6.3   | 6.6   | 5.4   | 4.8   | 4.7   | 4.8   | 5.0   | 5.7   | 6.4   |
|                                                                        | 5% WB       | MCDWR  | 17.8  | 14.6  | 12.8  | 11.3  | 11.1  | 10.8  | 8.8   | 9.4   | 10.1  | 10.8  | 12.0  | 15.0  |
|                                                                        |             | MCWBR  | 8.7   | 7.8   | 7.1   | 6.0   | 6.5   | 5.8   | 5.0   | 4.7   | 4.9   | 5.0   | 5.8   | 6.9   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.516 | 0.516 | 0.589 | 0.618 | 0.638 | 0.592 | 0.513 | 0.495 | 0.492 | 0.494 | 0.496 | 0.504 |
|                                                                        | taud        |        | 1.906 | 1.950 | 1.831 | 1.830 | 1.831 | 1.957 | 2.199 | 2.260 | 2.250 | 2.193 | 2.072 | 1.973 |
|                                                                        | Ebn at Noon |        | 234   | 246   | 234   | 230   | 224   | 232   | 252   | 257   | 256   | 250   | 239   | 232   |
|                                                                        | Edh at Noon |        | 56    | 57    | 67    | 68    | 68    | 59    | 47    | 44    | 44    | 44    | 48    | 51    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1424  | 1675  | 1854  | 1882  | 1693  | 1397  | 1399  | 1471  | 1474  | 1458  | 1451  | 1330  |
|                                                                        | RadStd      |        | 68    | 83    | 106   | 145   | 185   | 151   | 156   | 110   | 84    | 74    | 91    | 75    |

## Historical Trends

|                            | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only          | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page